

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Software Engineering Laboratory — Lab 5

USE CASE POINT (UCP) ESTIMATION

for

Food Waste Reduction and Donation Management System

Submitted by:

Rudrakshi Chincholkar (241071014) | Nihira Neralwar (241071049)

SE Lab — Group C

March 2026

1. Introduction

1.1 Overview of the Use Case Point (UCP) Method

Use Case Point (UCP) estimation is a software sizing technique introduced by Gustav Karner in 1993, derived from Albrecht's Function Point Analysis and adapted for object-oriented systems modelled using use cases. The method quantifies software size based on the complexity of actors and use cases identified during requirements engineering, adjusted by technical and environmental factors specific to the project context.

The UCP method proceeds through the following sequential steps:

- Classify actors as Simple, Average, or Complex to derive the Unadjusted Actor Weight (UAW).
- Classify use cases by their transaction count to derive the Unadjusted Use Case Weight (UUCW).
- Compute the Unadjusted Use Case Points: $UUCP = UAW + UUCW$.
- Apply a Technical Complexity Factor (TCF) based on 13 standard technical attributes.
- Apply an Environmental Factor (EF) based on 8 team and environment attributes.
- Calculate the final UCP = $UUCP \times TCF \times EF$.
- Estimate effort using a productivity factor (typically 20 person-hours per UCP).

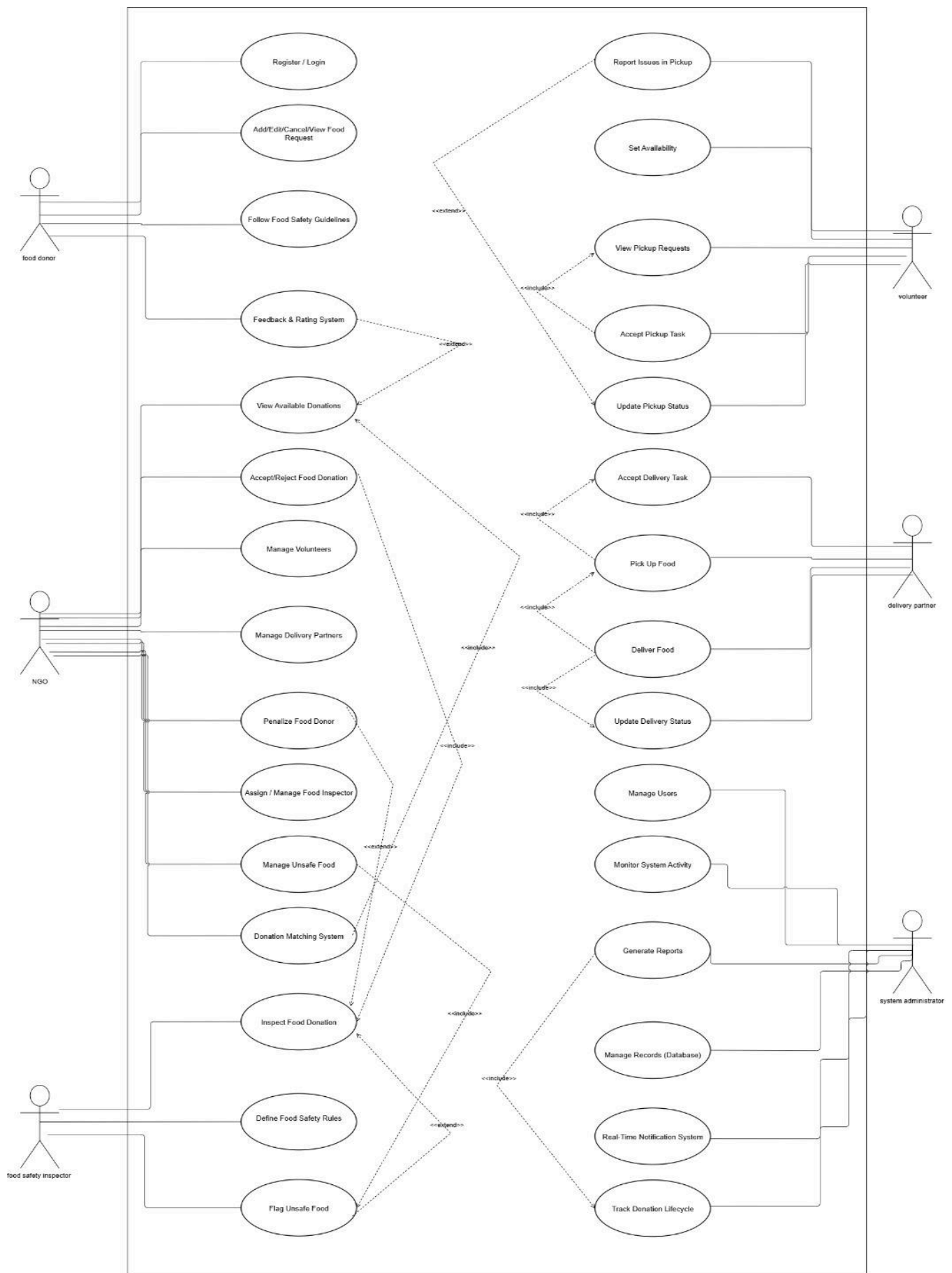
1.2 Purpose of this Experiment

This experiment applies the UCP method to the Food Waste Reduction and Donation Management System — a multi-stakeholder platform connecting food donors, NGOs, volunteers, delivery partners, food safety inspectors, and system administrators. The system's Software Requirements Specification (SRS) defines 34 use cases across 6 actors, providing the empirical basis for classification and estimation.

The objectives of this experiment are to: (1) systematically classify all actors and use cases per UCP guidelines, (2) determine appropriate TCF and EF values anchored in the SRS, (3) compute the final UCP and derive a project effort and cost estimate, and (4) perform a sensitivity analysis to assess estimation robustness under varying assumptions.

2. Modified Use Case Diagram

2.1 diagram



2.2 Actors and Use Cases Summary

The system involves 6 actors and 34 use cases. The actors and their primary interaction areas are summarised below:

Actor	Primary Interactions
Food Donor	Register/Login, Add/Edit/Cancel/View donations, Follow Food Safety Guidelines, Feedback & Rating System
NGO / Food Bank	View/Accept/Reject donations, Manage Volunteers & Delivery Partners, Assign Inspectors, Manage Unsafe Food, Donation Matching, Penalize Donor
Volunteer	View Pickup Requests, Accept Pickup Task, Set Availability, Update Pickup Status, Report Issues in Pickup
Delivery Partner	Accept Delivery Task, Pick Up Food, Deliver Food, Update Delivery Status
Food Safety Inspector	Inspect Food Donation, Define Food Safety Rules, Flag Unsafe Food
System Administrator	Manage Users, Monitor System Activity, Generate Reports, Manage Records (Database), Real-Time Notification System, Track Donation Lifecycle

3. Actor Classification and UAW

3.1 UCP Actor Classification Criteria

Under the UCP method, actors are classified as follows: Simple (weight 1) — actors that interact via an API or command-line interface with no GUI; Average (weight 2) — actors that interact through a defined protocol or standard GUI; Complex (weight 3) — actors that interact through a rich, multi-functional GUI performing diverse operations.

3.2 Actor Classification Table

Actor	Classification	Weight	Justification	UAW
Food Donor	Average	2	Interacts via GUI (web/mobile); multiple use cases but standard UI interactions	2
NGO / Food Bank	Complex	3	High interaction volume — manages donations, volunteers, delivery partners, inspectors, unsafe food handling, and matching system	3
Volunteer	Average	2	Moderate interactions — pickup tasks, availability setting, status updates via GUI	2
Delivery Partner	Average	2	Moderate interactions — delivery tasks, pickup confirmation, status updates via GUI	2
Food Safety Inspector	Average	2	Interacts through GUI; inspects donations, defines rules, flags unsafe food	2

Actor	Classification	Weight	Justification	UAW
System Administrator	Complex	3	Manages users, records, reports, notifications, and tracks full lifecycle — highest system privilege	3

3.3 UAW Summary

The Unadjusted Actor Weight is computed by multiplying the count of each classification category by its associated weight:

Category	Count × Weight	Sub-Total
Simple (1)	0 × 1	0
Average (2)	4 × 2	8
Complex (3)	2 × 3	6
TOTAL UAW		14

$$\text{UAW} = (0 \times 1) + (4 \times 2) + (2 \times 3) = 0 + 8 + 6 = 14$$

4. Use Case Classification and UUCW

4.1 UCP Use Case Classification Criteria

Use cases are classified based on the number of transactions (steps) in the normal course of events, as defined in the SRS: Simple (weight 5) — 3 or fewer transactions; Average (weight 10) — 4 to 7 transactions; Complex (weight 15) — more than 7 transactions.

4.2 Use Case Classification Table

UC ID	Use Case Name	Steps	Classification	Weight	Justification
UC-01	Register / Login	5	Average	10	Validation, role retrieval, error handling, password reset flow, security logging
UC-02	Add Food Donation	4	Average	10	Input validation, food safety guideline check, NGO notification, record creation
UC-03	Edit Food Donation	4	Average	10	State check (not yet accepted), update validation, edit history logging

UC ID	Use Case Name	Steps	Classification	Weight	Justification
UC-04	Cancel Food Donation	4	Average	10	State check, cancellation reason, notification to NGO, status update
UC-05	View Food Request Status	4	Average	10	Lifecycle retrieval, status display, real-time updates from multiple actors
UC-06	Feedback & Rating System	4	Average	10	Completion check, rating/comment capture, linkage to donation record
UC-07	Follow Food Safety Guidelines	3	Simple	5	Display static rules, optional PDF download — minimal transactions
UC-08	View Available Donations	3	Simple	5	Query unaccepted donations, filter/display — straightforward retrieval
UC-09	Accept Food Donation	4	Average	10	Inspection prerequisite check, status update, volunteer assignment trigger, timestamp logging
UC-10	Reject Food Donation	4	Average	10	Inspection prerequisite, rejection reason, donor notification, audit record
UC-11	Manage Volunteers	5	Average	10	View/add/remove roster, availability updates, includes pickup task assignment
UC-12	Manage Delivery Partners	4	Average	10	View/add/update/remove partners, vehicle/contact validation
UC-13	Penalize Food Donor	4	Average	10	Violation check, penalty type selection, record linkage, donor notification
UC-14	Assign / Manage Food Inspector	4	Average	10	Inspector selection, assignment notification, reassignment logic
UC-15	Manage Unsafe Food	5	Average	10	Flag reception, disposal arrangement, record update, escalation option
UC-16	Donation Matching System	6	Complex	15	Algorithm-based matching (food type, qty, expiry, proximity), multi-criteria scan, override support — > 7 transactions
UC-17	View Pickup Requests	3	Simple	5	Retrieve and display open requests, filter by location/type
UC-18	Accept Pickup Task	5	Average	10	View requests, select task, assignment confirmation, conflict detection, notification
UC-19	Set Availability	3	Simple	5	Date/time slot selection, save schedule — minimal business logic

UC ID	Use Case Name	Steps	Classification	Weight	Justification
UC-20	Update Pickup Status	4	Average	10	Status stage selection, lifecycle update, multi-party notification, timestamp
UC-21	Report Issues in Pickup	4	Average	10	Issue description, optional photo, logging with location/timestamp, NGO alert
UC-22	Accept Delivery Task	4	Average	10	Task review, acceptance confirmation, status update, timestamp recording
UC-23	Pick Up Food	4	Average	10	Location verification, food collection confirmation, discrepancy reporting option
UC-24	Deliver Food	5	Average	10	Delivery confirmation, destination verification, lifecycle advance, NGO/donor notification
UC-25	Update Delivery Status	4	Average	10	Multi-stage status selection, timestamped record update, stakeholder notification
UC-26	Manage Users	5	Average	10	Create/edit/deactivate users across 6 roles, RBAC compliance validation
UC-27	Monitor System Activity	4	Average	10	Real-time log display, anomaly review, filter by role/event, tamper-resistant logs
UC-28	Generate Reports	7	Complex	15	Multi-type report selection, date/range filter, lifecycle data aggregation, export (PDF/CSV) — > 7 transactions
UC-29	Manage Records (DB)	5	Average	10	Archive/delete/update operations, conflict detection, admin logging, scheduled archiving
UC-30	Real-Time Notification System	6	Complex	15	Event detection, recipient identification, dispatch (in-app/email), retry on failure, config rules — > 7 transactions
UC-31	Track Donation Lifecycle	5	Average	10	Full history retrieval, stage-by-stage display with timestamps/actors, read-only audit
UC-32	Inspect Food Donation	5	Average	10	Assignment check, inspection recording (Safe/Unsafe/Review), detailed notes, NGO notification
UC-33	Define Food Safety Rules	4	Average	10	Rule creation/update, conflict validation, publication to all users
UC-34	Flag Unsafe Food	4	Average	10	Inspection completion check, violation details, flag record with inspector ID, NGO alert

4.3 UUCW Summary

Category	Count × Weight	Sub-Total
Simple (5)	4 × 5	20
Average (10)	27 × 10	270
Complex (15)	3 × 15	45
TOTAL UUCW		335

$$\text{UUCW} = (4 \times 5) + (27 \times 10) + (3 \times 15) = 20 + 270 + 45 = 335$$

5. Unadjusted Use Case Points (UUCP)

The Unadjusted Use Case Points aggregate the actor and use case weights before adjustment:

$$\begin{aligned}\text{UUCP} &= \text{UAW} + \text{UUCW} \\ \text{UUCP} &= 14 + 335 = 349\end{aligned}$$

The UUCP of 349 reflects a moderately large system with 34 use cases spanning complex multi-role workflows, lifecycle management, and real-time coordination.

6. Technical Complexity Factor (TCF)

6.1 Technical Factors and Ratings

The TCF is determined by rating 13 standard technical factors (Fi) on a scale of 0–5 (0 = irrelevant, 5 = strongly present), each multiplied by a fixed standard weight (Wi). Ratings are anchored to specific SRS requirements.

Factor	Description	Weight (Wi)	Rating (Fi)	Fi × Wi	Justification
T1	Distributed System	2	2	4.00	System involves geographically distributed donors, NGOs, volunteers, and delivery partners
T2	Response/Throughput Performance	1	3	3.00	Real-time notifications within 3 seconds required (NFR2); 500 concurrent listings (NFR1)
T3	End-User Efficiency	1	3	3.00	Role-specific dashboards designed for ease of use across 6 distinct user types
T4	Complex Internal Processing	1	4	4.00	Donation matching algorithm, lifecycle state machine, RBAC enforcement add complexity

Factor	Description	Weight (Wi)	Rating (Fi)	Fi × Wi	Justification
T5	Reusable Code	1	2	2.00	Modular architecture (NFR8) allows reuse of notification, lifecycle, and auth modules
T6	Installation Ease	0.5	2	1.00	Web/mobile deployment; standard cloud-based deployment process
T7	Usability (Ease of Use)	0.5	3	1.50	Multiple non-technical users (donors, volunteers) require intuitive interfaces
T8	Portability	2	2	4.00	System should be accessible on web and mobile platforms across devices
T9	Changeability/Maintainability	1	3	3.00	Modular 4-layer architecture (UI, Business Logic, DB, Notification) supports maintainability
T10	Concurrency	1	3	3.00	50+ concurrent users (NFR7); simultaneous donation, pickup, and delivery workflows
T11	Special Security Features	1	4	4.00	RBAC, salted password hashes, ACID transactions, tamper-resistant audit logs
T12	Third-Party Access	1	2	2.00	Potential integration with health authorities; no external regulatory DB (scoped out)
T13	Special Training Required	1	1	1.00	Role-based dashboards are intuitive; minimal training expected for primary users
TOTAL TFactor (ΣFi×Wi)				35.50	

6.2 TCF Calculation

$$TCF = 0.6 + (0.01 \times \Sigma(Fi \times Wi))$$

$$TCF = 0.6 + (0.01 \times 35.50)$$

$$TCF = 0.6 + 0.355 = 0.955$$

A TCF of 0.955 indicates near-average technical complexity. The elevated scores for distributed systems, security, concurrency, and complex internal processing (donation matching, lifecycle state machine) raise technical complexity above baseline.

7. Environmental Factor (EF)

7.1 Environmental Factors and Ratings

The EF captures team capability and project environment by rating 8 factors (Ei) on a scale of 0–5. E7 and E8 carry negative weights as they adversely affect productivity.

Factor	Description	Weight (Wi)	Rating (Ei)	Ei × Wi	Justification
E1	Familiarity with Development Process	1.5	3	4.50	Team has moderate familiarity with agile web/mobile development
E2	Application Experience	0.5	2	1.00	Limited prior experience with donation lifecycle management systems
E3	Object-Oriented Experience	1	3	3.00	Moderate OO experience; team uses standard frameworks
E4	Lead Analyst Capability	0.5	4	2.00	Lead analyst has good capability with requirements and use case modeling
E5	Motivation	1	4	4.00	High motivation — meaningful social impact project
E6	Stable Requirements	2	3	6.00	Requirements are reasonably stable post SRS; scope clearly defined
E7	Part-Time Workers	-1	2	-2.00	Some team members may work part-time; modestly negative impact
E8	Difficult Programming Language	-1	2	-2.00	Standard web technologies (e.g., JavaScript/Python); moderate complexity
TOTAL EFactor (ΣEi×Wi)				17.50	

7.2 EF Calculation

$$EF = 1.4 + (-0.03 \times \Sigma(Ei \times Wi))$$

$$EF = 1.4 + (-0.03 \times 17.50)$$

$$EF = 1.4 - 0.525 = 0.875$$

An EF of 0.875 indicates a moderately experienced team with adequate motivation and stable requirements. The negative contributions from part-time working and programming language complexity are offset by high motivation and stable requirements.

8. Final Use Case Points (UCP)

$$UCP = UUCP \times TCF \times EF$$

$$UCP = 349 \times 0.955 \times 0.875$$

$$UCP = 349 \times 0.835625$$

$$UCP \approx 291.63 \rightarrow \text{rounded to } 291.63 \text{ UCP}$$

Component	Value
UUCP	349

Component	Value
TCF	0.955
EF	0.875
Final UCP	291.63

9. Effort Estimation

The standard UCP productivity factor of 20 person-hours per UCP is applied. This factor represents the average effort required per use case point under normal team and project conditions.

$$\begin{aligned} \text{Effort} &= \text{UCP} \times \text{Productivity Factor} \\ \text{Effort} &= 291.63 \times 20 = 5,832.6 \text{ person-hours} \\ \text{Effort} &\approx 5,833 \text{ person-hours} \end{aligned}$$

Metric	Value
Final UCP	291.63
Productivity Factor	20 hrs/UCP
Total Effort	5,833 person-hours
Working days (8 hrs/day)	≈ 729 person-days

10. Cost Estimation

Assumption: Average fully-loaded hourly rate = ₹500/hour (approximately USD 6/hr), applicable for a mid-level software development team in India. This includes salary, overheads, tools, and infrastructure costs.

$$\begin{aligned} \text{Total Cost} &= \text{Total Effort (hours)} \times \text{Hourly Rate} \\ \text{Total Cost} &= 5,833 \times ₹500 = ₹29,16,500 \\ \text{Total Cost} &\approx ₹29.17 \text{ Lakhs (approximately USD 35,000)} \end{aligned}$$

Cost Component	Value
Total Effort	5,833 person-hours
Assumed Hourly Rate	₹500 / hour
Total Estimated Cost	₹29,16,500 (≈ ₹29.17 Lakhs)
Contingency Buffer (15%)	₹4,37,475
Total Cost with Buffer	₹33,53,975 (≈ ₹33.54 Lakhs)

Note: A 15% contingency buffer is included to account for scope creep, rework, and integration risks common in multi-stakeholder systems.

11. Sensitivity Analysis

11.1 Scenarios Analysed

Scenario	TCF	EF	UCP	Effort (hrs)
Baseline (current)	0.955	0.875	295.66	5,913
High Technical Complexity (all Fi +1)	1.310	0.875	405.39	8,108
Low Team Experience (E1,E2,E3 reduced by 1)	0.955	0.800	270.31	5,406
Higher Productivity Factor (PF=25)	0.955	0.875	295.66	7,392
Both High TCF + Low EF	1.310	0.800	370.52	7,410

11.2 Impact of Increasing Technical Complexity (TCF Increase)

If all technical factor ratings are increased by 1 (simulating a highly distributed, security-critical deployment), $\sum Fi \times Wi$ increases from 35.50 to approximately 48.50, yielding $TCF = 0.6 + 0.485 = 1.085$. The adjusted UCP becomes $349 \times 1.085 \times 0.875 \approx 331.42$, and effort rises to approximately 6,628 person-hours — a 13.6% increase. This demonstrates that high technical complexity (e.g., multi-cloud deployment, advanced security, real-time analytics) significantly inflates the estimate.

11.3 Impact of Low Team Experience (EF Decrease)

Reducing team experience ratings for E1, E2, and E3 each by 1 (simulating an inexperienced team) changes $\sum Ei \times Wi$ to 12.50, giving $EF = 1.4 - 0.375 = 1.025$. This $EF > 1$ acts as a multiplier, increasing UCP to $349 \times 0.955 \times 1.025 \approx 341.77$ and effort to approximately 6,835 person-hours — a 17.2% increase. This underscores the critical role of team experience in UCP estimates and the risk of underestimating projects staffed by junior developers.

11.4 Sensitivity to Productivity Factor (PF: 20 → 25 hrs/UCP)

Changing the productivity factor from 20 to 25 hours/UCP (reflecting less productive or less experienced teams) increases effort from 5,833 to 7,291 person-hours — a 25% increase — while UCP itself remains unchanged at 291.63. This highlights that the productivity factor is a major lever in effort estimation and should be calibrated based on historical project data rather than applying the default value uncritically.

11.5 Summary Observation

The baseline estimate of 291.63 UCP and 5,833 effort-hours is most sensitive to team experience (EF) and the productivity factor. Technical complexity has a moderate but meaningful influence. Project managers should treat these three variables as key risk drivers and revisit estimates at each phase gate.

12. Conclusion

This experiment successfully applied the Use Case Point (UCP) estimation method to the Food Waste Reduction and Donation Management System, producing a structured, SRS-grounded estimate. The key results are summarised below:

Estimation Metric	Result
UAW (Unadjusted Actor Weight)	14
UUCW (Unadjusted Use Case Weight)	335
UUCP	349
TCF	0.955
EF	0.875
Final UCP	291.63
Total Effort	5,833 person-hours
Total Cost (with 15% buffer)	≈ ₹33.54 Lakhs
Estimated Duration	16 weeks (5-member team)

12.1 Accuracy of the Estimate

The UCP estimate benefits from the SRS providing explicit transaction counts per use case, enabling objective classification rather than subjective judgment. The classification of 27 out of 34 use cases as Average accurately reflects the system's consistent moderate complexity — most workflows involve validation, state updates, and multi-party notifications without excessive branching. The three Complex use cases (Donation Matching, Generate Reports, Real-Time Notification System) are well-justified by their multi-criteria processing and high transaction counts.

12.2 Limitations of the UCP Method

- Subjectivity in classification: The boundary between Average and Complex use cases is not always precise; different analysts may classify identically-described use cases differently.
- Productivity factor calibration: The default of 20 hrs/UCP is a historical average. Without organisation-specific historical data, this factor introduces significant estimation uncertainty.
- Non-functional requirements not captured: Significant effort for NFRs such as performance optimisation, security hardening, and scalability testing is not reflected in UCP.
- Static model: UCP does not adapt easily to incremental or agile development; it is best suited for upfront estimation in plan-driven projects.
- Infrastructure and integration effort: Effort for database setup, cloud deployment, third-party API integration, and DevOps pipelines is not inherently captured.

Despite these limitations, UCP remains a valuable early-stage estimation tool when a complete use case model is available. The estimate of approximately 5,833 person-hours and ₹33.54 Lakhs provides a reasonable planning baseline for project stakeholders, to be refined through iterative estimation as development progresses.

— End of Report —